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The Palaeomagnetism of the Kisii Series, Western Kenya

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Summary

The Precambrian Kisii Series in Western Kenya is unmetamorphosed and contains andesites, rhyolites, quartzites and basalts. The basalts, for which a minimum K–Ar age of 930 My is cited, were sampled at 15 sites, and after A.F. treatment nine sites yielded a stable palaeomagnetic direction which is believed to be primary. The mean direction ($N = 9$) after tilt correction is $D = 99°$, $I = -59°$, $\alpha_{95} = 10°$, and the corresponding palaeomagnetic south pole is at 6° N, 12° W with $A_{95} = 14°$. This pole lies close to several other poles for Africa but it is of a very different age and it cannot be correlated with them. Instead it is suggested that the Kisii pole must form part of a previously undefined polar wander path for the Late Precambrian of Africa proposed in an accompanying paper by Brock and Piper.

1. Introduction

The palaeomagnetic record in East Africa is concentrated mainly in the Tertiary where the extensive outcrop of volcanics associated with the rift valleys has encouraged detailed sampling. Elsewhere in Africa, however, palaeomagnetic work has yielded a number of results from the Precambrian. It is therefore of interest to consider the possibility of obtaining Precambrian results from East Africa. The oldest of the Precambrian systems in Kenya are the Nyanzian and Kavirondian, both older than 2200 My (Pulfrey 1969). Although not highly metamorphosed these systems have suffered intensive folding and shearing, and are unsuitable for palaeomagnetic work. They are overlain by the unmetamorphosed and relatively undeformed Kisii Series which is the subject of this study. The Kisii Series has been placed in the Late Precambrian, partly because of its undeformed state as contrasted with the older Nyanzian–Kavirondian, and partly because it has been correlated on lithological grounds with the Bukoban of Tanzania and Uganda. Consistent with this is the single K–Ar age of 675 ± 25 My reported by Cahen & Snelling (1966).

2. Geology and age

The Kisii Series occurs in the Nyanza Province, western Kenya, where it forms the Kisii Highlands. The adjacent lower country is mainly formed of Nyanzian and Kavirondian rocks. Three main divisions have been recognized (Shackleton 1946): (1) Basalts; (2) Quartzites; (3) Felsites, andesites and rhyolites.

A detailed description is given by Huddleston (1951) who shows that the Series has a shallow basin like structure with dips seldom greater than 10°. The structure is best seen in the quartzite which to the west forms a prominent scarp. Underlying the quartzite are the basalts, having an average thickness of about 300 m, and exposed

in a belt over a width of about 8 km to the west of the quartzite scarp. The lavas are mainly non-porphyritic and appear in hand specimen with little variation as fine grained dense dark blue-grey or dark green rocks with occasional vesicles. In composition the feldspars are most commonly oligoclase and the pyroxenes diopsidic augite. Magnetite and sphene are abundant accessories. Sampling was confined to the basalts, and Fig. 1 is a geological sketch map showing the location of the sampling sites.

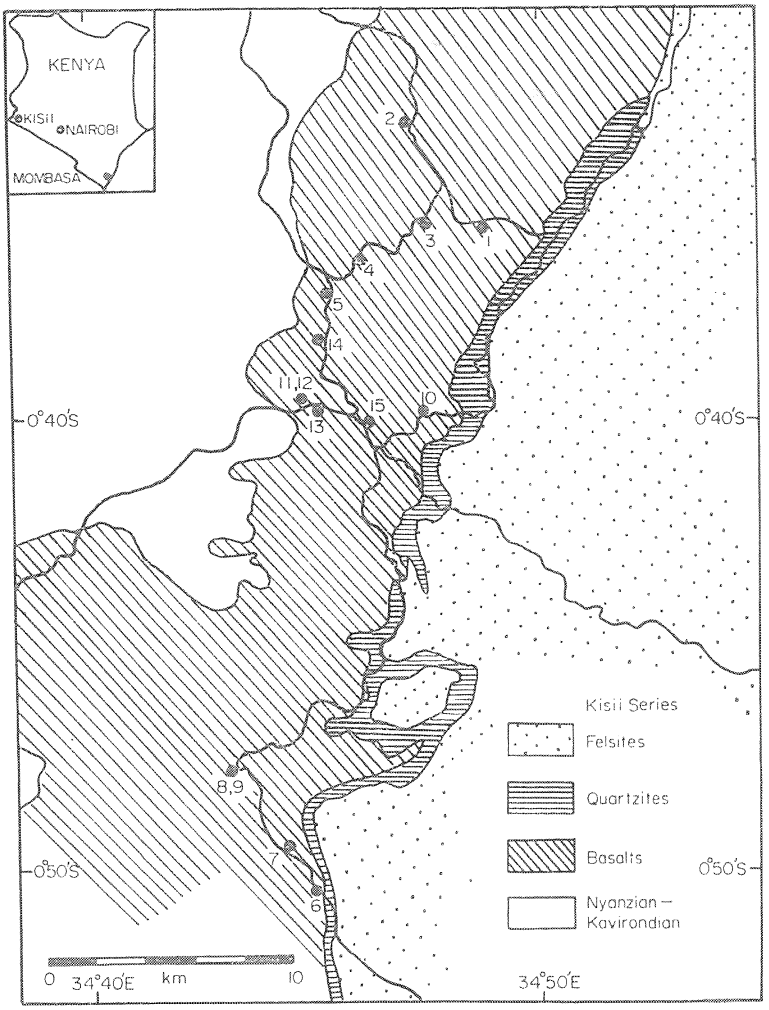
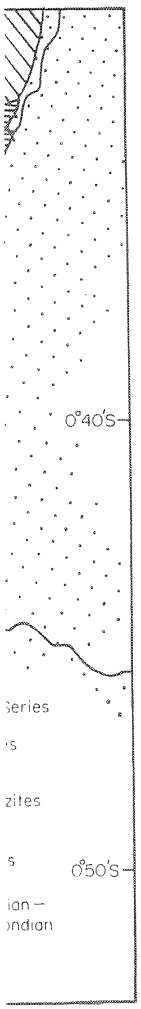


FIG. 1. Geological sketch map showing location of sites.

Samples from two of the palaeomagnetic sites gave whole rock K–Ar ages of 906 ± 35 My and 964 ± 35 My (Sites 9 and 15 respectively). These determinations were kindly carried out by Dr D. C. Rex and Mrs J. Gronow of the Leeds University Institute of African Geology and are reported by Briden *et al.* (1971). The ages differ by rather more than their errors and until a larger number of fully concordant ages are available it is probably wise to regard these determinations as indicating a minimum age of about 930 My. This is greater than the previously quoted age of 675 My which must be assumed to have been affected by argon loss. It is also greater than the age of about 800 My quoted by Piper (1972) for the intrusives and lavas of the Bukoban System in Tanzania. The correlation of the Kisii Series with the

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Bukoban System may therefore have to be abandoned. The problem of the relationship between the Bukoban System and the Kisii Series is further considered in an accompanying paper (Brock & Piper 1972) where it is assessed in the light of palaeomagnetic, stratigraphic, and radiometric evidence.

3. Field-work and measurement

A total of 116 cores of diameter 2·2 cm were collected from 15 sites using a small hand-held petrol driven drill of the type described by Doell & Cox (1967). Sites were found in quarries (3), in or near river beds (7), and on hillsides (5). The dissected nature of the country made site selection fairly easy, and outcrop, although not extensive, was usually fresh. Orientation was by magnetic compass and precautions were taken to avoid errors due to small regions of intensely magnetized rock. The cores were cut into specimens 2·5 cm long for measurement.

Measurements were made with an astatic magnetometer having a maximum sensitivity of $1{\cdot}3 \times 10^{-5}$ emu of moment per mm deflection which enabled intensities of about 4 or 5×10^{-5} emu cm^{-3} to be measured. Later a spinner magnetometer of the air turbine type described by Gough (1964) was developed by one of us (J.B.V.) for routine measurement down to intensities of about 5×10^{-7} emu cm^{-3}. This provided sufficient sensitivity for all samples encountered in this study.

The isolation of a stable component from the initial NRM required the extensive use of the usual alternating field demagnetizing procedures. Initially one or two specimens from each site were subjected to step-wise partial demagnetization in an attempt to find a suitable cleaning field for the remainder of the specimens from the site. In some sites a single field suitable for all specimens could not be found, and most specimens then required full step-by-step demagnetization before an end-direction could be identified. In addition it was necessary to discard a number of specimens which behaved erratically. This type of behaviour is not uncommon in pre-Tertiary rocks, and it was in anticipation of such problems that a fairly large number of samples were collected from each site.

In the analysis, samples were given unit weight and where several specimens had been cut from one core these were averaged to yield a single sample direction for use in the statistical analysis. Site mean directions were then given unit weight in the calculation of the required formation means. This procedure conforms to the conservative approach outlined by Irving (1964, p. 64). A two-tier analysis was not attempted since there was a considerable variation in within-site scatter.

4. Results

Of the 15 sites, most showed very scattered NRM directions, and 6 were random at the 95 per cent level by the Watson (1956) criterion. With hindsight it is possible to discern a slight clustering of the NRM directions in the region of the final results, but there is certainly no obvious suggestion of the cleaned direction such as can often be seen in the NRM directions of younger rocks. Paradoxically the scattered nature of the NRM was of assistance in judging the success of the cleaning procedures since it made movement into tighter groups more evident. This can be seen in the tabulated results where three of the random sites became well grouped after cleaning (Table 1), and six other sites showed significant increases in the value of k, the estimate of Fisher's precision parameter. The remaining six sites showed no clear end direction even after cleaning. In five of these cases the directions remained scattered throughout the demagnetization procedure with individual samples moving erratically in unstable fashion. Such sites could not be used and were discarded. The other unsuccessful site (site 5), however, showed great stability. The NRM directions were random,

Table 1

Site mean directions and poles

	Total NRM *In situ*					After A.F. cleaning *In situ*								After tilt correction			
Site	N	R	k	D	I	H_{AF}	N	R	k	D	I	Lat.	Long.	D	I	Lat.	Long.
1	9	4·90	2	127	−59	350	8	7·70	23	94	−51	3° S	157° E	88	−58	2° N	163° E
2	9	4·70	—	Random		300	9	8·83	47	122	−37	30° S	149° E	112	−46	28° S	157° E
3	4	2·90	—	Random		—	—	—	—	—	—	—	—	—	—	—	—
4	6	3·49	—	Random		300	6	5·44	9	102	−53	9° S	159° E	92	−61	1° S	168° E
5	7	3·51	—	Random		—	—	—	—	—	—	—	—	—	—	—	—
6	5	3·09	—	Random		—	—	—	—	—	—	—	—	—	—	—	—
7	9	7·07	4	51	−52	400	6	5·60	12	101	−53	8° S	159° E	103	−57	10° S	163° E
8	6	4·27	3	10	−12	—	—	—	—	—	—	—	—	—	—	—	—
9	7	6·92	74	53	−80	300	7	6·97	228	104	−80	4° S	196° E	117	−85	4° S	206° E
10	11	8·00	3	23	−7	400	6	5·64	14	100	−36	9° S	145° E	97	−44	2° S	151° E
11	9	8·09	9	92	−20	150	8	7·78	32	96	−30	5° S	141° E	94	−37	4° S	145° E
12	7	4·24	2	119	−3	—	—	—	—	—	—	—	—	—	—	—	—
13	9	3·68	—	Random		400	5	4·87	31	106	−54	13° S	160° E	101	−62	8° S	169° E
14	7	4·82	3	55	−44	—	—	—	—	—	—	—	—	—	—	—	—
15	11	9·57	7	48	−73	300	8	7·89	63	93	−71	1° S	180° E	76	−78	1° N	192° E

N Number of samples
R Resultant of N unit vectors
k Estimate of Fisher's precision parameter
D, I Declination and inclination in degrees
H_{AF} Peak cleaning field, Oersted
Lat., Long. Latitude and Longitude of virtual geomagnetic north pole, degrees

but d magnetization in fields as high as 1000 Oe produced almost no change in the individual directions and reduced intensities by only about 30 per cent. A higher cleaning field might have yielded a result, but such fields were not available and the site had to be discarded.

Of the nine successful sites, three were clear cut (Sites 2, 4 and 9), all samples giving well-grouped directions after demagnetization. In the remaining cases it was necessary to discard one or more of the samples before a good group could be obtained. The rejected samples were of two types. One type, which occurred in sites 10, 13 and 15, showed erratic behaviour during demagnetization and never yielded a stable component. The other type gave an apparently stable direction lying well outside the group formed by the remaining samples, typically by angles between 50° and 80°. Such directions were rejected on the ground that they did not belong to the population being sampled; the argument being that in each case the divergent direction differed from the group mean by more than the angle θ_{95} ($= 140/k^{\frac{1}{2}}$ degrees) which is the radius of the circle enclosing 95 per cent of individual directions. Sites 1, 7 and 11 contained samples of this type. Site 11 showed an unusual behaviour. In fields up to 150 Oe an initially rather loose group becomes progressively tighter, but in fields from 200 Oe upwards individual samples began to behave erratically moving through large angles with each demagnetization step. This behaviour could be due to the presence of a finite minimum intensity which continued demagnetization randomizes but does not remove (Irving 1964, p. 93). The cleaned direction was taken at the field of tightest grouping.

The site data is given in Table 1, where the cleaned directions are shown both before and after tilt correction. Tilt is obtained from values given for the overlying quartzite by Huddleston (1951). The corrections are similar for all sites, and no significant improvement in grouping results, but since the tilt is tectonic rather than depositional the tilt corrected values are preferred. The main effect is to increase the inclination by about 8°.

It is believed that these cleaned directions represent a primary component of magnetization for the following reasons. The directions are well removed from any recent field direction, and indeed from any post-Cambrian direction for Africa. They do however, coincide with a late Precambrian–Cambrian direction, and an ancient secondary magnetization might be suspected. But there is no evidence either in the geology or in the K–Ar ages for any significant reheating at this time. In particular, there is no evidence that the Pan-African thermal event, so pervasive elsewhere in Africa, has affected the Kisii Series. An ancient secondary magnetization due to this event is therefore unlikely. Finally, the improvement in grouping after cleaning of both sample directions and of site directions, and the movement of both towards a consistent set of cleaned directions is strong evidence for an underlying stable primary direction. This will always remain a strong argument in palaeomagnetism, and is one of the reasons for attempting a wide spread of sites and for collecting a reasonable number of samples from each site, when planning field work.

Site mean directions are illustrated in Fig. 2 in which there is a suggestion that the distribution is not circular, the directions having a greater spread in inclination than in declination. This could be due to unsuspected variation of tilt, to real polar movement, to insufficiently averaged secular variation, or to inadequately cleaned secondary components. The number and spread of sites, and the experimental procedures make the last two possibilities unlikely. The presence of real polar movement would be surprising but not impossible. Unsuspected tilts, however, must be accepted as a real possibility, but, by definition, no correction can be made for them. An overall mean has been calculated from these directions by two methods, one in which a single pole with an oval of confidence is obtained from the mean of site directions, and the other in which a mean pole with a circle of confidence is obtained from the site poles. These means, before and after tilt correction, are shown in Table 2.

R Resultant of N unit vectors
k Estimate of Fisher's precision parameter
D, I Declination and inclination in degrees
H_{AF} Peak cleaning field, Oersted
Lat., Long. Latitude and Longitude of virtual geomagnetic north pole, degrees

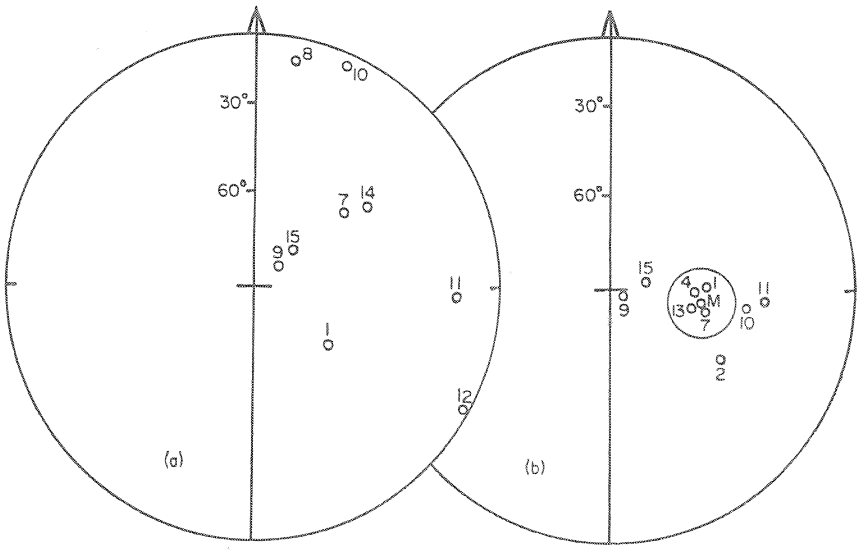


FIG. 2. (a) Site mean directions of NRM. Only those sites which are significant at the 95 per cent level by the Watson (1956) criterion are shown. (b) Significant site mean directions after A.F. cleaning and tilt correction. The mean is denoted by M and the 95 per cent circle of confidence is shown.

The mean pole after tilt correction is the preferred result, and for comparison with other African results it is shown in Fig. 3 as a south pole lying off the coast of West Africa at 6° N, 12° W with a 95 per cent circle of confidence of 14°. It is the main outcome of this study.

5. Discussion

The African polar wander path is fully described in the review paper by McElhinny *et al.* (1968), and in this section we shall retain their pole notation (P1–P12 for the Precambrian and B1–B19 for the Phanerozoic). There are several other pole positions near the Kisii pole. One is a Late Precambrian–Cambrian group represented by poles B1 and B2. The presumed Late Precambrian age for the Kisii Series would have been consistent with these pole positions, and it was the expectation that such a pole might be found which motivated this investigation. However the radiometric work mentioned earlier rules out such a possibility and sets an age for the Kisii Series of at least 930 My. The African polar wander path has another very much older pole in the same region. It is the pole P7 from the Mashonaland dolerites in Rhodesia which are now believed to have an age of between 1850 and 1900 My (W. Compston and M. W. McElhinny, in preparation). The K–Ar age for Kisii is probably a minimum age, but it is very unlikely that the result underestimates the age so severely that the Kisii Series can be correlated with the Mashonaland dolerites. If the Kisii age is taken at its face value at about 930 My, the nearest other African poles in age are that for the Pilansberg dykes (P12) at about 1300 My, and those from the Bukoban intrusives and lavas of Tanzania at about 800 My (Piper 1972). All these poles are well removed from the Kisii pole. Thus we have the situation where those poles to which the Kisii pole is most closely related in space are well removed in time, and those which are probably closest to it in time are distant in space. We must conclude therefore that the Kisii pole position is *sui generis*, and that the pole for Africa has been in the vicinity of west Africa on at least three

Table 2

Formation mean directions and poles after A.F. cleaning

	Direction					Palaeomagnetic north pole					
	N	D	I	k	α_{95}	Lat.	Long.	K	A_{95}	dp	dm
Uncorrected	9	103	−52	22	11	9° S	161° E	18	13		
						1° S	158° E			10	15
Tilt corrected	9	99	−59	24	11	6° S	168° E	15	14		
						7° S	165° E			12	16

N Number of sites
D, I Declination and inclinations, degrees
k, K Estimate of Fisher's precision parameter
α_{95}, A_{95} Radius of 95 per cent circle of confidence, degrees
dp, dm Semi-axes of 95 per cent oval of confidence, degrees

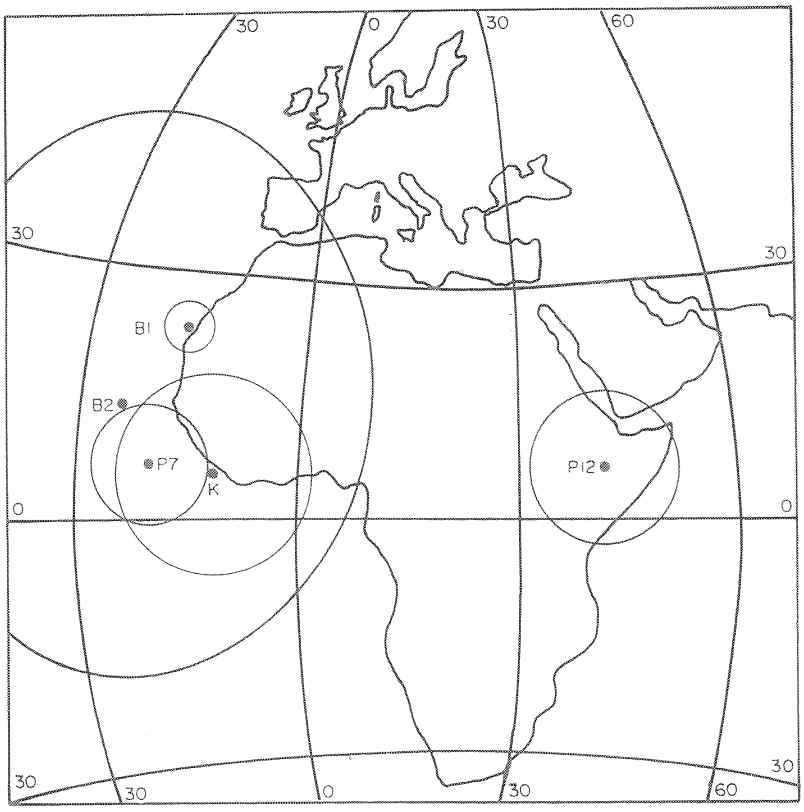


FIG. 3. Map showing pole positions and circles of confidence. K, Kisii Series (this study). P12, Pilansberg dykes, P7, Mashonaland dolerites, B1, Ntonya ring complex, B2, Hook intrusives (from McElhinny *et al.* (1968)).

separate occasions in the past. The polar wander curve, however, must be continuous and must link the separate poles in some way. In particular we can hope that a fairly direct path will link the Pilansberg pole through Kisii to the Bukoban poles, spanning the interval 1300 to 800 My. A proposal for such a curve is made in an accompanying paper (Brock & Piper 1972).

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